Diagonalization

Q: Let $D=\begin{bmatrix}5 & 0\\ 0 & 3\end{bmatrix}$, $D^3 = ?$ $\begin{bmatrix}5^3 & 0\\ 0 & 3^3\end{bmatrix}$

Q: Let $A=\begin{bmatrix}1 & 3\\ 0 & 2\end{bmatrix}$, $A=PDP^{-1}=\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & 2\end{bmatrix}\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}^{-1}$, $A^3 = ?$

$$\begin{aligned}A^3 &= (PDP^{-1})(PDP^{-1})(PDP^{-1})\\ &= PDP^{-1}PDP^{-1}PDP^{-1}\\ &= PDDDP^{-1}\\ &= PD^3P^{-1}\\ &= \begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}\begin{bmatrix}1^3 & 0\\ 0 & 2^3\end{bmatrix}\begin{bmatrix}1 & 3\\ 0 & 1\end{bmatrix}^{-1}\end{aligned}$$

The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A. In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P.

$$\underset{n\times n}{A} = \underbrace{\begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}}_{P}\underbrace{\begin{bmatrix}\lambda_1 & & 0\\ & \ddots & \\ 0 & & \lambda_n\end{bmatrix}}_{D}\underbrace{\begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix}^{-1}}_{P^{-1}}$$

If A is diagonalizable, to find diagonalization:

1. Find all eigenvalues & corresponding eigenvectors
2. Place eigenvalues on the diagonals of D and eigenvectors in cols of P

Diagonalizability:

A diagonalizable $\Leftrightarrow$ A has n linearly independent eigenvectors

(When algebraic multiplicities = geometric multiplicities)

Always true: geometric multiplicities $\leq$ algebraic multiplicities $= n$

Example:

Let $A = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & 5 \end{bmatrix}$ eigenvalues : $\underbrace{3, 3}_{\text{algebraic multiplicity}=2}, 5 \leftarrow$ algebraic multiplicity $= 1$.

$\Rightarrow (3-\lambda)^2 (5-\lambda)^1 = 0$

characteristic equation: $\det(A-\lambda I)=0 \Rightarrow \det\left(\begin{bmatrix} 3-\lambda & 1 & 2 \\ 0 & 3-\lambda & 3 \\ 0 & 0 & 5-\lambda \end{bmatrix}\right) = 0$

λ-eigenspace $= \text{Nul}(A-\lambda I)$ # free variables in $A-\lambda I = \dim \text{Nul}(A-\lambda I)$

$$A-3I = \begin{array}{c} \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} \\ \begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ \text{free} & \text{pivots} & \end{array} \end{array}$$

$\dim \text{Nul}(A-3I) = \dim$ 3-eigenspace $= \underline{1} \leftarrow$ geometric multiplicity $= 1$

$= \#$ L.I. eigenvectors

$$A-5I = \begin{array}{c} \begin{bmatrix} -2 & 1 & 2 \\ 0 & -2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \\ \begin{array}{ccc} \uparrow & \uparrow & \uparrow \\ \text{pivots} & & \text{free} \end{array} \end{array}$$

$\dim \text{Nul}(A-5I) = \dim$ 5-eigenspace $= \underline{1} \leftarrow$ geometric multiplicity $= 2$

$= \#$ L.I. eigenvectors

Geometric Multiplicities < Algebraic Multiplicities

$2+1 < 1+1$

Practice Problem

Show that if a and b are real, then the eigenvalues of $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ are $a \pm bi$, with corresponding eigenvectors $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ i \end{bmatrix}$.

Complex Eigenvalues

Ex: Let $A = \begin{bmatrix} a & -b \\ b & a \end{bmatrix} \Rightarrow \det(A - \lambda I) = \lambda^2 - tr(A)\lambda + \det(A)$

$$= \lambda^2 - 2a\lambda + (a^2+b^2) = 0$$

$$\lambda = \frac{2a \pm \sqrt{4a^2 - 4(a^2+b^2)}}{2}$$

$$\lambda = \frac{2a \pm \sqrt{4a^2 - 4a^2 - 4b^2}}{2}$$

$$\lambda = \frac{2a \pm 2\sqrt{-b^2}}{2} = a \pm bi$$

eigenvectors: $A - (a+bi)I = \begin{bmatrix} -bi & -b \\ b & -bi \end{bmatrix} \sim \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix}$

$x_1 = ix_2 \Rightarrow \vec{x} = x_2 \begin{bmatrix} i \\ 1 \end{bmatrix} = c \boxed{\begin{bmatrix} 1 \\ -i \end{bmatrix}}$

$A - (a-bi)I = \begin{bmatrix} bi & -b \\ b & bi \end{bmatrix} \sim \begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix}$

$\vec{x} = x_2 \begin{bmatrix} -i \\ 1 \end{bmatrix} = c \boxed{\begin{bmatrix} 1 \\ i \end{bmatrix}}$

EXAMPLE 6 If $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, where a and b are real and not both zero, then the eigenvalues of C are $\lambda = a \pm bi$. (See the Practice Problem at the end of this section.) Also, if $r = |\lambda| = \sqrt{a^2 + b^2}$, then

$$C = r\begin{bmatrix} a/r & -b/r \\ b/r & a/r \end{bmatrix} = \overbrace{\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix}}^{\text{scaling}} \overbrace{\begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}}^{\text{rotation}}$$

where φ is the angle between the positive x-axis and the ray from $(0, 0)$ through (a, b). See Figure 2 and Appendix B. The angle φ is called the *argument* of $\lambda = a + bi$. Thus

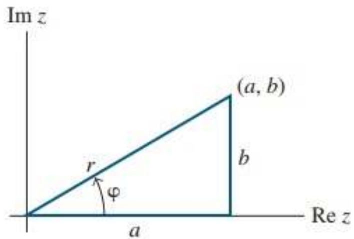


FIGURE 2

Let $\lambda = a + bi$, $\underbrace{\bar{\lambda}}_{\text{conjugate}} = a - bi$

Let $Ax = \lambda x$

then $A\bar{x} = \overline{A}\,\overline{x} = \overline{Ax} = \overline{\lambda x} = \bar{\lambda}\bar{x}$

Hence $A\bar{x} = \bar{\lambda}\bar{x}$

Thus if λ is an eigenvalue of A, then $\bar{\lambda}$ is also an eigenvalue of A

Inner Product, Length, and Orthogonality

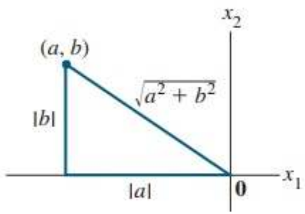
Inner Product ($\vec{u} \cdot \vec{v}$)

$$\vec{u} \cdot \vec{v} = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Length ($\|\vec{u}\|$)

The **length** (or **norm**) of $\mathbf{v}$ is the nonnegative scalar $\|\mathbf{v}\|$ defined by

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}, \quad \text{and} \quad \|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$$



Orthogonality

Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are **orthogonal** (to each other) if $\mathbf{u} \cdot \mathbf{v} = 0$.

Orthogonal Component ($W^\perp$)

To provide practice using inner products, we introduce a concept here that will be of use in Section 6.3 and elsewhere in the chapter. If a vector $\mathbf{z}$ is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then $\mathbf{z}$ is said to be **orthogonal to** W. The set of all vectors $\mathbf{z}$ that are orthogonal to W is called the **orthogonal complement** of W and is denoted by $W^\perp$ (and read as "W perpendicular" or simply "W perp").

$$W^\perp = \{ z \in \mathbb{R}^n \mid z \cdot w = 0 \quad \forall\, w \in W \}$$

Let A be an $m \times n$ matrix. The orthogonal complement of the row space of A is the null space of A, and the orthogonal complement of the column space of A is the null space of A^T:

$$(\text{Row } A)^\perp = \text{Nul } A \quad \text{and} \quad (\text{Col } A)^\perp = \text{Nul } A^T$$

Orthogonal Sets

Let $S = \{u_1, u_2, \cdots, u_p\}$ is a set of orthogonal vectors, called an orthogonal set.

An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y}$ in W, the weights in the linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y}\cdot\mathbf{u}_j}{\mathbf{u}_j\cdot\mathbf{u}_j} \qquad (j = 1, \ldots, p)$$

Orthogonal Projection

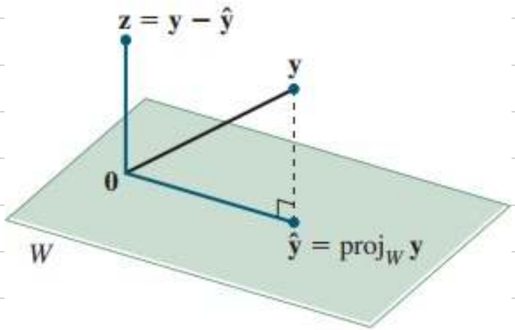


FIGURE 2

Finding α to make $\mathbf{y} - \hat{\mathbf{y}}$ orthogonal to $\mathbf{u}$.

$$\boxed{\hat{\mathbf{y}} = \mathrm{proj}_L \mathbf{y} = \frac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u}}$$

Orthonormal Set

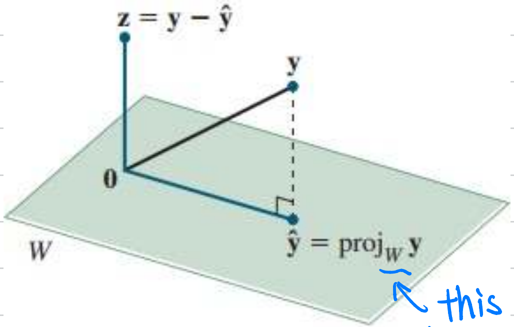
- Orthogonal Set
- All vectors in the set have unit length (length = 1)

An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I$.

Orthogonal Projections

Recall from last time:

Orthogonal Projection



this is when W is 1-D

FIGURE 2
Finding α to make $\mathbf{y} - \hat{\mathbf{y}}$ orthogonal to $\mathbf{u}$.

$$\boxed{\hat{\mathbf{y}} = \text{proj}_L \mathbf{y} = \frac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u}}$$

The Orthogonal Decomposition Theorem

Let W be a subspace of $\mathbb{R}^n$. Then each $\mathbf{y}$ in $\mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} \tag{1}$$

where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$. In fact, if $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is any orthogonal basis of W, then

$$\hat{\mathbf{y}} = \frac{\mathbf{y}\cdot\mathbf{u}_1}{\mathbf{u}_1\cdot\mathbf{u}_1}\mathbf{u}_1 + \cdots + \frac{\mathbf{y}\cdot\mathbf{u}_p}{\mathbf{u}_p\cdot\mathbf{u}_p}\mathbf{u}_p \tag{2}$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

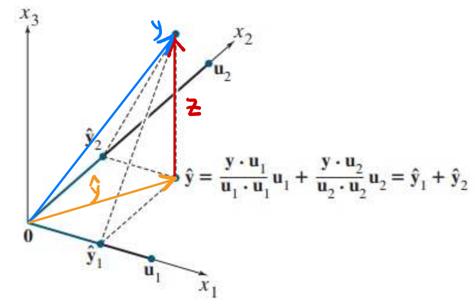


FIGURE 3 The orthogonal projection of $\mathbf{y}$ is the sum of its projections onto one-dimensional subspaces that are mutually orthogonal.

Gram-Schmidt Process

The Gram–Schmidt Process

Given a basis $\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$
\begin{aligned}
\mathbf{v}_1 &= \mathbf{x}_1 \\
\mathbf{v}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 \\
\mathbf{v}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 \\
&\vdots \\
\mathbf{v}_p &= \mathbf{x}_p - \frac{\mathbf{x}_p \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_p \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2}\mathbf{v}_2 - \cdots - \frac{\mathbf{x}_p \cdot \mathbf{v}_{p-1}}{\mathbf{v}_{p-1} \cdot \mathbf{v}_{p-1}}\mathbf{v}_{p-1}
\end{aligned}
$$

Then $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is an orthogonal basis for W. In addition

$$\operatorname{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \operatorname{Span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} \qquad \text{for } 1 \le k \le p \tag{1}$$

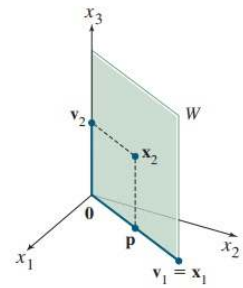


FIGURE 1
Construction of an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$.

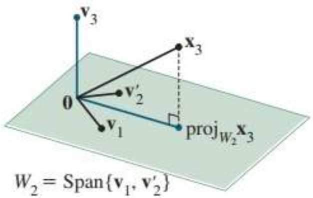


FIGURE 2 The construction of $\mathbf{v}_3$ from $\mathbf{x}_3$ and W_2.

QR Factorization

The QR Factorization

If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as $A = QR$, where Q is an $m \times n$ matrix whose columns form an orthonormal basis for Col A and R is an $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.

- columns of Q can be find by the Gram-Schmidt Process, then normalize
- Since columns of Q are orthogonal, then $Q^T = Q^{-1}$, so that $R = Q^T A$.

Least-Squares Problems

If A is $m \times n$ and $\mathbf{b}$ is in $\mathbb{R}^m$, a **least-squares solution** of $A\mathbf{x} = \mathbf{b}$ is an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ such that

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| \leq \|\mathbf{b} - A\mathbf{x}\|$$

for all $\mathbf{x}$ in $\mathbb{R}^n$.

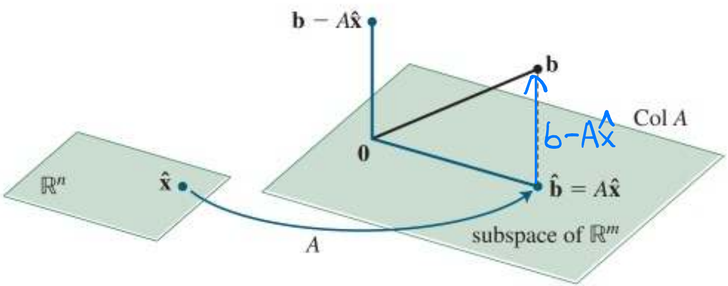


FIGURE 2 The least-squares solution $\hat{\mathbf{x}}$ is in $\mathbb{R}^n$.

The least-squares solution $\hat{x}$ is obtain when $\hat{b} = A\hat{x}$ is the projection of $\hat{b}$ onto Col A

The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ coincides with the nonempty set of solutions of the normal equations $A^TA\mathbf{x} = A^T\mathbf{b}$.

$$A^T(b - A\hat{x}) = 0 \Rightarrow A^Tb - A^TA\hat{x} = 0 \Rightarrow A^TA\hat{x} = A^Tb$$

Let A be an $m \times n$ matrix. The following statements are logically equivalent:

a. The equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution for each $\mathbf{b}$ in $\mathbb{R}^m$.

b. The columns of A are linearly independent.

c. The matrix A^TA is invertible.

When these statements are true, the least-squares solution $\hat{\mathbf{x}}$ is given by

$$\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b} \tag{4}$$

$$A\hat{x} = QR\hat{x} = QRR^{-1}Q^Tb$$
$$= QQ^Tb = \text{Proj}_{\text{Col}Q}\, b = \text{Proj}_{\text{Col}A}\, b = \hat{b}$$

Given an $m \times n$ matrix A with linearly independent columns, let $A = QR$ be a QR factorization of A as in Theorem 12. Then, for each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a unique least-squares solution, given by

$$\hat{\mathbf{x}} = R^{-1}Q^T\mathbf{b} \tag{6}$$